

GAYATRI SHARMA KURMATEY

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PROFESSIONAL SUMMARY

Data Engineer and AI/ML Engineer with 6+ years of experience building scalable data pipelines, cloud platforms, streaming workflows, analytics systems, automation solutions, and machine learning applications across automotive, energy, logistics, education, supply chain, and research. Skilled in SQL, Python, AWS, Kafka, Azure Synapse, PostgreSQL, BI, machine learning, deep learning, and Generative AI. Currently developing VoltX, an EV charging intelligence platform on the Tesla Fleet API Developer platform using Python, FastAPI, React, OAuth 2.0, geospatial logic, and event-driven notifications.

SKILLS

Automation & Workflow: Airflow, Power Apps, Power Automate, Power Pages, Dataverse, Event-Driven Notifications, Microsoft Copilot, ChatGPT, Cursor, Codex, Claude

BI & Data Visualization: Power BI, Tableau, Qlik Sense, Google Analytics, Matplotlib, Seaborn, Visio, Chimera

Cloud & Infrastructure: AWS, Amazon S3, Amazon SageMaker, Azure, Google Cloud, Microsoft Intune, SharePoint

APIs & Application Development: Tesla Fleet API Developer Platform, REST APIs, OAuth 2.0, FastAPI, React, Geospatial Logic, Browser & Mobile Notifications, Event-Driven Architecture

Data Formats: JSON, CSV, XLSX, SQL, PDF, GPickle, FASTA, PDB, PNG, JPEG

Data Science: BioPython, Jupyter Notebook, KarateClub, NetworkX, Python, R, SAS, Scikit-Learn, TensorFlow, PyTorch

Databases, Warehousing & ETL: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, Sybase, SQLdbx, BigQuery, Snowflake, Amazon Redshift, AWS Glue, Azure Data Factory, SSIS, dbt, ETL/ELT Pipelines, Data Modeling, Data Warehousing, Apache Kafka

DevOps & Containers: CI/CD, Git, GitHub, Version Control, Testing, Debugging, Code Review, Docker, Kubernetes, Linux, Ubuntu/WSL

Excel & Scripting: Advanced Excel, Macros, Pivot Tables, VBA, Advanced Functions, Google Sheets

Experimentation & Analysis: A/B Testing, Case Study Testing, Experimental Design, Hypothesis Testing, Statistical Modeling

Financial Analytics: AML, Fraud Detection, KPI Modeling, Risk Dashboards, Transaction Monitoring, FP&A

Governance & Enterprise Systems: Data Governance, Access Control, RBAC, Data Privacy, API Security, Secrets Management

Machine Learning, Deep Learning & Generative AI: LLMs, BERT, Transformers, Foundation Models, CNNs, Mask R-CNN, Graph Neural Networks, Geometric Deep Learning, Semantic Modeling, SageMaker

Project & Team Management: Confluence, JIRA, Microsoft Project, Monday.com

Reporting & Business Analytics: KPI Dashboards, Executive Reporting, Operational Dashboards, Weekly/Monthly Business Reporting, Data Storytelling, DAX, Power Query

Trust, Safety & Compliance: Role-Based Security, Policy Enforcement, Risk Monitoring, Data Privacy

WORK EXPERIENCE

Belong Automotive Technologies, LLC, Texas, *Project Lead Data Engineer*

Oct 2025 – Jul 2026

- Led 2 interns in the development of Python ETL/ELT pipelines, PostgreSQL data models, and Odoo ERP and PLM workflows across 50+ supply-chain functions, including inventory, purchasing, suppliers, demand planning, product data, and operations
- Integrated ERP and PLM data across inventory, purchasing, supplier, BOM, and product lifecycle workflows, building analytics-ready data models and improving cross-functional visibility across supply-chain operations
- Partnered with China and U.S. operations teams to translate inventory, purchasing, supplier, and demand-planning requirements into scalable ERP workflows, data models, and reporting assets, improving global supply-chain coordination and decision-making
- Reduced manual reporting effort by 12 hours/week and improved KPI visibility from 9 days to 2 hours through automated reporting workflows, saving 600+ analyst hours annually and an estimated \$40K–\$75K in productivity value
- Accelerated U.S. launch readiness by securing the company domain, designing and deploying the official U.S. website, setting up Microsoft 365, and creating communication workflows for 100+ team members in 8 weeks

SkyTech Inc, New Jersey, *Data Analyst (Contractor)*

Oct 2024 – Sep 2025

- Tested HR workflow automation configurations, improving dashboard adoption by 25%
- Integrated HR data across MSSQL, APIs, Tableau, and Power BI, improving data interoperability by 30% and enabling faster workforce reporting across compliance, HR operations, and leadership dashboards
- Automated RBAC and compliance workflows in Azure Synapse, strengthening governance controls, reducing manual access-review effort by 15+ hours/week, and lowering operational risk across HR data processes

Robert Half, Fremont Unified School District, California, *Data Science Engineer (Contractor)*

Mar 2024 – Jan 2025

(via SkyTech Inc from Oct 2024)

- Led a 6-member team to build SQL, Azure Synapse, and Power Platform ETL pipelines for student database migration, saving 25 hours/week in manual processing and improving reliability of district-level reporting operations
- Built Power BI access-control dashboards for IT teams to track faculty laptop activations and inactive users, reducing activation follow-up time by 15 hours/week
- Automated inclusion, diversity, and workforce equity reporting for 3,360+ students, improving student data validation accuracy by 30%
- Streamlined ERP-integrated business process automation for 2,500+ asset and HR workflows, improving KPI tracking, forecasting accuracy, and data reliability across finance, IT, and workforce operations
- Coached cross-functional teams on Power Platform and Tableau dashboards, driving 100% adoption within one week and saving 18 hours/week in recurring manual reporting effort

Pacific Gas and Electric Company, California, *Business Intelligence Analyst (Contractor)*

May 2023 – Aug 2023

- Optimized Spark SQL data lake and ETL pipelines for operational dashboards, reducing refresh time from 60 minutes to 20 minutes and improving timely access to business-critical utility metrics

- Automated SQL-to-SharePoint reporting workflows using Python, Tableau, and Power BI, saving 20+ hours/week and reducing recurring manual reporting costs for operational analytics teams
- Trained analysts on Power Query, DAX, and Tableau calculations, reducing recurring dashboard support requests by 30% and saving 10+ hours/week

Tesla Inc, California, Data Science Engineer Intern

May 2022 – Dec 2022

- Developed space planning and people analytics dashboards using Power Platform and Dynamics 365, saving 10+ hours weekly
- Created 20+ Power BI and Tableau dashboards for PMs and executives monitoring \$1.3B+ CAPEX projects, improving portfolio visibility and reducing report refresh time by 40 minutes daily through DAX optimization
- Built S-curve forecasting tool for 500+ construction projects, applying statistical testing and predictive analytics to support \$1.3B+ CAPEX planning, improve logistics and financial forecasting, and save 20+ hours/week
- Enforced role-based access controls via Azure AD and SharePoint, strengthening data governance and ensuring compliance across Power Platform systems

Tango Analytics, Texas, Data Scientist Intern

Jan 2022 – May 2022

- Engineered deep-learning computer vision pipelines in a Linux environment using Python, OpenCV, and Mask R-CNN to detect and annotate floor-plan objects, improving dataset quality for downstream ML workflows
- Cleaned, labeled, and validated image datasets for entity recognition models, and semantic segmentation, achieving 88% accuracy and reducing production integration effort by 65%

Amazon, India, Associate Logistics Data Analyst

Aug 2018 – Aug 2019

- Improved last-mile logistics analytics by uncovering customer delivery insights, boosting delivery efficiency by 80%, resolving 100+ operational issues, and saving 110+ hours/month across logistics support workflows
- Analyzed Amazon Flex pilot feedback and delivery performance data to improve onboarding, training, software workflows, and escalation resolution across U.S.–India logistics operations

Indian Space Research Organization, India, Data Science Intern

May 2017 – Mar 2018

- Built a materialized-view procurement analytics system using multidimensional data modeling, Sybase/MySQL, SQLdbx, and COWAA integration, with an HTML/CSS/Java interface to visualize query results and support procurement decision-making
- Developed a conference hall booking system using Sybase, MySQL, HTML, CSS, and Java, enabling ISRO employees to manage room reservations, prevent scheduling conflicts, and improve booking efficiency by 99%

RESEARCH & PROJECTS

VoltX EV Charging Intelligence Agent, Self Project, [Demo](#)

Sep 2026 – Present

- Built a full-stack EV charging intelligence platform using the Tesla Fleet API Developer platform to retrieve vehicle state and nearby Superchargers
- Developed geospatial logic with proactive browser and mobile alerts, reducing the need to manually check charging rates or repeatedly open the Tesla app
- Implemented Tesla OAuth 2.0 with server-side credential handling; extending VoltX with historical rate analysis, cost optimization, and predictive charging recommendations

NASA Ames Research Center, California, Open-Source Researcher

Feb 2024 – Present

- Built atomic-level protein graph representations with BioPython/NetworkX and evaluated Graph2Vec + Weisfeiler-Lehman embeddings to compare active and inactive kinase conformations
- Applied PCA, t-SNE, and KMeans to assess embedding separability and explored geometric GNNs, SE(3)-equivariant transformers, and SO(3)-equivariant autoencoders for improved protein structure representation

Jet Propulsion Laboratory, NASA, California, Volunteer Research Assistant

May 2023 – Aug 2023

- Compiled Generative AI, LLMs, BERT, and Haystack for semantic modeling, knowledge extraction, and advanced R&D analytics

The University of Texas at Dallas, Research Assistant

Aug 2021 – May 2022

- Developed a containerized Open Data Cube environment on Linux/Ubuntu using Docker and AWS S3, automating ingestion of 500+ NASA satellite images for scalable storage and analysis
- Tested ML models (SVM, Neural Networks, Logistic Regression, and Stacking) with feature selection to predict road accident severity

EDUCATION

Master of Science - Information Technology & Management, University of Texas at Dallas

Jan 2021 – Dec 2022

Bachelor of Technology - Computer Science, Jawaharlal Nehru Technological University

Sep 2014 – Jun 2018

HONORS & AWARDS

- Recognized by superintendents, district officials and Vice Presidents for automation and dashboard initiatives at Tesla and FUSD
- Scholar with Recognition – Ranked Top 3 GPA in MS-ITM Class, University of Texas at Dallas

CERTIFICATES

Semantic Segmentation with Amazon SageMaker with Amazon AWS

May 2026

Introduction to Large Language Models (LLM)

May 2026

Introduction to Generative AI

May 2026

Applying AI Principles with Google Cloud

May 2026

Databricks Generative AI Foundation

Jun 2026

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Dec 2025